



UNIVERSITÀ DI PISA

Mott localisation, antiferromagnetism and nematic superconductivity in the extended Hubbard model with non-local pairing

Candidato:

Alessandro Gori

Relatore:

Prof. Giacomo Mazza

Outline

1. Cuprates and Hubbard physics
2. The Hartree-Fock method

Cuprates and Hubbard physics

Nobel Prize in Physics 1987



Photo from the Nobel Foundation
archive.

J. Georg Bednorz

Photo share: 1/2



Photo from the Nobel Foundation
archive.

K. Alexander Müller

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Possible High T_c Superconductivity in the Ba–La–Cu–O System

J.G. Bednorz and K.A. Müller

IBM Zürich Research Laboratory, Rüschlikon, Switzerland

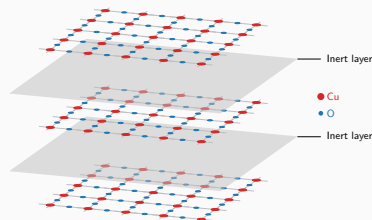
Received April 17, 1986

What makes cuprates interesting?

- ◇ Strong “unconventional” insulators;
- ◇ Rich phase diagram;
- ◇ Origin of Cooper pairing: still unclear;
- ◇ Highest critical temperatures for SC transition in this class of materials.

These materials are stackings of:

- ◇ 2D copper oxide active layers;
- ◇ intermediate layers of inert material.



Most of the cuprates physics is well described by the single-band Hubbard model on the square lattice:

$$\hat{H}_{\text{HM}} = \underbrace{-t \sum_{\langle ij \rangle} \sum_{\sigma} \hat{c}_{i\sigma}^{\dagger} \hat{c}_{j\sigma}}_{\text{Kinetic operator}} + \underbrace{U \sum_i \hat{n}_{i\uparrow} \hat{n}_{i\downarrow}}_{\text{On-site repulsion}} \quad \text{with } t, U > 0.$$

The phase diagram of cuprates

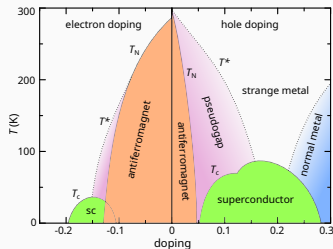
Cuprates exhibit various phases, among which:

- ◇ **Antiferromagnetic long-range order** at null and low doping;
- ◇ ***d*-wave superconductivity** as we inject or remove charge.

At **half-filling**, band theory would predict a metallic state. Instead we observe strong insulation.



Cuprates are **not** band insulators.



Schematic phase diagram of cuprates.
Author: Holger Motzkau, [Wikimedia commons](#), CC BY-SA 3.0.

Mott localisation: a many-body insulator

Band theory assumption: e - e interactions are **negligible**.

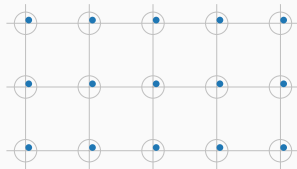
Strongly coupled Hubbard model:
on-site repulsion

$$U \sum_i \hat{n}_{i\uparrow} \hat{n}_{i\downarrow},$$

much bigger than kinetics. **Big energetic cost** of having two electrons on the same site.



Insulation mechanism: electrons are **localised** as in a sort of “traffic jam”.



Mott localisation: a many-body insulator

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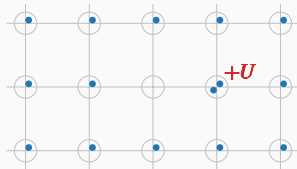
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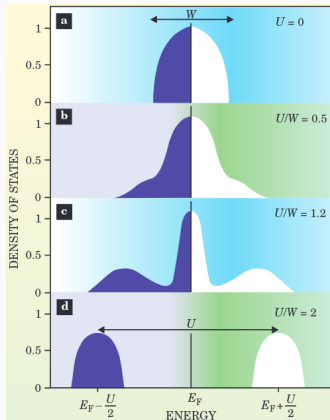
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Bands flattening: a signature of Mott localisation



- (a) Free model, conventional DoS;
- (b) Weak coupling, large coherent peak;
- (c) Intermediate coupling, **narrow coherent peak**;
- (d) Strong coupling, lower and upper Hubbard bands.



Localisation = flattening of bare bands.

Gabriel Kotliar and Dieter Vollhardt, 2004.
Physics Today 57 (March): 53–59.

How does Mott localisation relate to antiferromagnetism and superconductivity?

Antiferromagnetism

Perturbation theory at strong coupling gives:

$$\hat{H}_{\text{HM}} \simeq J \sum_{\langle ij \rangle} \hat{\mathbf{S}}_i \cdot \hat{\mathbf{S}}_j + \Delta E,$$

with antiferromagnetic exchange
 $J = 4t^2/U$.

“Mottness”: the ground manifold is localised.

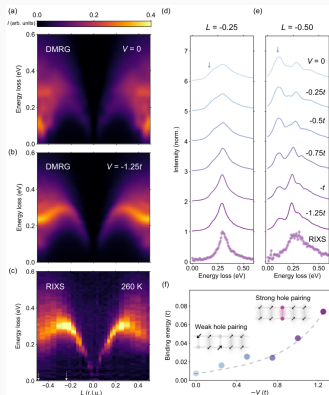
Superconductivity

Phenomenology: we observe *d*-wave superconductivity, i.e. the superconducting gap is anisotropic under $\pi/2$ rotations $\mathcal{R}$,

$$\begin{aligned} \left\langle \hat{c}_{\uparrow \mathbf{r}}^\dagger \hat{c}_{\downarrow \mathbf{r} + \Delta \mathbf{r}}^\dagger \right\rangle \\ = - \left\langle \hat{c}_{\uparrow \mathcal{R} \mathbf{r}}^\dagger \hat{c}_{\downarrow \mathcal{R}(\mathbf{r} + \Delta \mathbf{r})}^\dagger \right\rangle. \end{aligned}$$

“Mottness”: non-local Cooper pairing.

Recent developments: the extended Hubbard model



Hari Padma et al., 2025. *Physical Review X* 15 (2): 021049.

Incompatibilities with the spectrum of magnetic excitation in cuprates.



Proposal: extension of the model with a **non-local attraction**,

$$\hat{H}_{\text{EHM}} = \hat{H}_{\text{HM}} - V \underbrace{\sum_{\langle ij \rangle} \hat{n}_i \hat{n}_j}_{\hat{H}_V}.$$

We know from BCS theory that $\hat{H}_V$ can act as a **source of d -wave superconductivity**.

The model (EHM):

$$\hat{H}_{\text{EHM}} = \underbrace{-t \sum_{\langle ij \rangle} \sum_{\sigma} \hat{c}_{i\sigma}^{\dagger} \hat{c}_{j\sigma}}_{\text{Kinetic operator}} + \underbrace{U \sum_i \hat{n}_{i\uparrow} \hat{n}_{i\downarrow}}_{\text{Local repulsion}} - \underbrace{V \sum_{\langle ij \rangle} \hat{n}_i \hat{n}_j}_{\text{Non-local attraction}} \quad \text{with } t, U, V > 0.$$

The motivation: Recent findings of compatibility with this model for experimental results.

The idea: we know that the EHM can host d -wave superconductivity, which is related to Mott physics. Can we obtain manifest signatures of Mott localisation in the ground-state?

The method: theoretical and numerical Hartree-Fock technique.

The Hartree-Fock method

We want to approximate the true ground state $|\Psi_0\rangle$ of the EHM, with energy E_0 .

Variational principle:

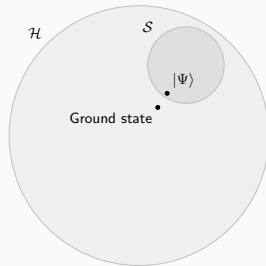
$$\forall |\Phi\rangle \in \mathcal{S} \subset \mathcal{H} : E[|\Phi\rangle] \geq E_0.$$

$\Downarrow$

Constrained minimisation procedure:

$$|\Psi\rangle : \left. \frac{\delta}{\delta |\Phi\rangle} E[|\Phi\rangle] \right|_{|\Phi\rangle=|\Psi\rangle} = 0.$$

The state $|\Psi\rangle$ is the best approximation to $|\Psi_0\rangle$ inside of $\mathcal{S}$.



Sketch of the Hilbert space $\mathcal{H}$ and a subset $\mathcal{S}$.

Consider all possible fermionic non-interacting hamiltonians:

$$\hat{H}_{\text{HF}} \equiv \sum_{\alpha} E_{\alpha} \hat{\gamma}_{\alpha}^{\dagger} \hat{\gamma}_{\alpha} \quad \text{with} \quad \left\{ \hat{\gamma}_{\alpha}, \hat{\gamma}_{\beta}^{\dagger} \right\} = \delta_{\alpha\beta},$$

with $\hat{\gamma} \leftrightarrow \hat{c}$ through a unitary map.

Hartree-Fock method: we identify $\mathcal{S}$ with the set of all the ground states of these hamiltonians,

$$\mathcal{S} = \left\{ |\Psi\rangle \mid |\Psi\rangle \text{ is the ground-state of some } \hat{H}_{\text{HF}} \right\}.$$

$\Downarrow$

Wick's theorem:

$$\langle \hat{c}_{\alpha}^{\dagger} \hat{c}_{\beta}^{\dagger} \hat{c}_{\gamma} \hat{c}_{\delta} \rangle = \underbrace{\langle \hat{c}_{\alpha}^{\dagger} \hat{c}_{\delta} \rangle \langle \hat{c}_{\beta}^{\dagger} \hat{c}_{\gamma} \rangle}_{\text{Hartree}} - \underbrace{\langle \hat{c}_{\alpha}^{\dagger} \hat{c}_{\gamma} \rangle \langle \hat{c}_{\beta}^{\dagger} \hat{c}_{\delta} \rangle}_{\text{Fock}} + \underbrace{\langle \hat{c}_{\alpha}^{\dagger} \hat{c}_{\beta}^{\dagger} \rangle \langle \hat{c}_{\gamma} \hat{c}_{\delta} \rangle}_{\text{Cooper}}.$$

Target parametric hamiltonian:

$$\hat{H}_{\text{HF}}[\mathbf{v}] \equiv \sum_{\alpha} E_{\alpha}[\mathbf{v}] \hat{\gamma}_{\alpha}^{\dagger}[\mathbf{v}] \hat{\gamma}_{\alpha}[\mathbf{v}] ,$$

being $\mathbf{v}$ a vector of variational parameters.

The entire variational problem can be reformulated equivalently:

Variational picture:

$$|\Psi_0\rangle \simeq |\Psi[\mathbf{v}]\rangle ,$$

where $\mathbf{v}$ is the vector of parameters that minimises energy

$$\left. \frac{\delta}{\delta |\Phi\rangle} E[|\Phi\rangle] \right|_{|\Phi\rangle=|\Psi[\mathbf{v}]\rangle} = 0 .$$

Self-consistency picture:

$$|\Psi_0\rangle \simeq |\Psi[\mathbf{v}]\rangle ,$$

where $\mathbf{v}$ is the vector of parameters that satisfies the self-consistency relation

$$\mathbf{A}(\mathbf{v}) = \mathbf{v} ,$$

being $\mathbf{A}$ a non-linear map that can be uniquely determined.

That's all, thank you!

Backup slides
